

Al-Khwarizmi’s method of solving equations

Al-jabr and al-muqabala — the two moves that gave algebra its name, written in Khiva twelve centuries ago.

LESSON 125–126

2 × 40 MIN

MATEMATIKA 7, §56

S8 2.5 HISTORICAL

LESSON CLOCK

10'

20'

20'

30'

5'

Who he was	10 min
Al-jabr — restoring	20 min
Al-muqabala — balancing	20 min
A medieval solution, and the modern one beside it	30 min
Homework	5 min

LEARNING OBJECTIVES

- Explain what al-jabr and al-muqabala mean and do.
- Solve an equation in the medieval style, without negative numbers.
- Connect the two moves to the modern rules for transposing and collecting.
- Place Al-Khwarizmi in the history of mathematics in Central Asia.

TEXTBOOK

National	pp. 371–377
Cambridge	Stage 8, pp. 31–36
Workbook	History sheet 2C

01

Explanation

Who he was

Muhammad ibn Musa **al-Khwarizmi** (c. 780 – c. 850) came from Khorezm, in what is now Uzbekistan, and worked at the House of Wisdom in Baghdad. His book *al-Kitab al-mukhtasar fi hisab al-jabr wa-l-muqabala* gave two words to the world.

WORD	FROM	MEANING TODAY
algebra	<i>al-jabr</i>	the whole subject
algorithm	his own name, Latinised	a finite list of steps

WHY HE WROTE IN WORDS, NOT SYMBOLS

There were no symbols for the unknown, no equals sign and no negative numbers. Every equation was a sentence, and every solution a paragraph. That is the difficulty his two moves were invented to overcome.

Al-jabr — restoring

Al-jabr means restoring: a subtracted quantity is removed from one side by adding it to both. It is how a negative term is got rid of when negative numbers do not exist.

$$5x - 3 = 12 \rightarrow 5x = 12 + 3$$

BEFORE	AFTER AL - JABR
$5x - 3 = 12$	$5x = 15$
$2x - 7 = x + 1$	$2x = x + 8$
$x^2 - 4x = 5$	$x^2 = 4x + 5$

THE SAME MOVE YOU ALREADY MAKE

Modern algebra says “transpose and change the sign”. Al-jabr says “add the missing amount to both sides”. They are the same operation; only the language differs.

Al-muqabala — balancing

Al-muqabala means balancing: like quantities on the two sides are cancelled against each other.

$$5x + 3 = 2x + 12 \rightarrow 3x = 9$$

BEFORE	AFTER AL - MUQABALA
$5x + 3 = 2x + 12$	$3x = 9$
$4x + 7 = x + 19$	$3x = 12$
$x^2 + 5 = 3x + 5$	$x^2 = 3x$

TWO MOVES, AND EVERY LINEAR EQUATION FALLS

Al-jabr removes what is subtracted; al-muqabala removes what is repeated. Applied in turn they reduce any linear equation to $ax = b$ — which is exactly the modern four-step method with older names.

A medieval solution, and the modern one beside it

AL - KHWARIZMI ' S WORDS	MODERN LINE
“Five things less three dirhams equal two things and nine dirhams.”	$5x - 3 = 2x + 9$
“Restore the three: five things equal two things and twelve dirhams.”	$5x = 2x + 12$
“Balance the two things: three things equal twelve dirhams.”	$3x = 12$
“Then one thing is four.”	$x = 4$

HIS TERM	OUR TERM
thing (shay)	x
dirham	the constant term
square (mal)	x^2
al-jabr	transposing a negative term
al-muqabala	collecting like terms

HE ALSO PROVED HIS METHODS GEOMETRICALLY

Where we write $(x + 3)^2 = x^2 + 6x + 9$, he drew the square and its four pieces — the same picture used two chapters ago. Without symbols, the diagram was the proof.

02 Worked examples

EXAMPLE 1 Solve $5x - 3 = 2x + 9$ in the medieval style, naming each move.

- ① Al-jabr: add 3 to both sides.

$$5x = 2x + 12$$

- ② Al-muqabala: cancel $2x$ from both sides.

$$3x = 12$$

- ③ Divide by 3.

$$x = 4$$

- ④ Check: $5 \cdot 4 - 3 = 17$ and $2 \cdot 4 + 9 = 17$ ✓

ANSWER

$$x = 4$$

EXAMPLE 2 Which move turns $4x + 7 = x + 19$ into $3x = 12$?

- ① The x terms are cancelled against each other.

- ② The dirhams are cancelled too.

- ③ That is al-muqabala, balancing.

ANSWER

Al-muqabala

EXAMPLE 3 Write “four things less two dirhams equal ten dirhams” as an equation and solve it.

① $4x - 2 = 10$

② Al-jabr: $4x = 12$.

③ $x = 3$

ANSWER

$x = 3$

03 Interactive model

Have the class solve one equation entirely in words, with no symbols at all; the effort involved explains better than any lecture why notation was worth inventing.

Which move is this?

$5x - 3 = 12$ becomes $5x = 15$ by:

al-jabr

al-muqabala

division

expanding

$5x + 3 = 2x + 12$ becomes $3x = 9$ by:

al-jabr

al-muqabala

division

squaring

The word “algebra” comes from:

al-muqabala

al-jabr

his name

Arabic for “number”

The word “algorithm” comes from:

al-jabr

al-muqabala

his name

Greek

He was born in:

Khorezm

Baghdad

Cairo

Samarkand

In his work, a “thing” means:

x^2

x

the constant

the answer

A “dirham” means:

x

x^2

the constant term

a fraction

He could not use:

addition

negative numbers

multiplication

fractions

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Al-Khwarizmi	Al-Xorazmiy	Аль-Хорезми
Al-jabr — restoring	Al-jabr — to'ldirish	Аль-джабр — восполнение
Al-muqabala — balancing	Al-muqobala — muvozanatlash	Аль-мукабала — сопоставление
Algebra	Algebra	Алгебра
Algorithm	Algoritm	Алгоритм
Manuscript	Qo'lyozma	Рукопись
House of Wisdom	Bayt ul-hikma	Дом мудрости
Heritage	Meros	Наследие

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Al-jabr means:

balancing

restoring

dividing

squaring

Al-muqabala means:

restoring

balancing

expanding

factorising

Al-Khwarizmi worked at:

the House of Wisdom

Oxford

Alexandria

Bukhara only

His century was the:

$2x - 7 = x + 1 \rightarrow 2x = x + 8$ is:

His proofs were:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Al-jabr means

ANSWER restoring

2. Al-muqabala means

ANSWER balancing

3. “Algebra” comes from

ANSWER al-jabr

4. “Algorithm” comes from

ANSWER his name

5. A “thing” is

ANSWER x

6. A “dirham” is

ANSWER the constant term

7. $5x - 3 = 12$ after al-jabr

ANSWER $5x = 15$

Medium · the standard question

7 PROBLEMS

1. $5x - 3 = 2x + 9$

ANSWER $x = 4$

2. $4x + 7 = x + 19$

ANSWER $x = 4$

3. “Four things less two dirhams equal ten dirhams”

ANSWER $4x - 2 = 10, x = 3$

4. $2x - 7 = x + 1$: which move first?

ANSWER Al-jabr

5. $5x + 3 = 2x + 12$: which move first?

ANSWER Al-muqabala

6. Why could he not write -3 ?

ANSWER Negative numbers were not in use

7. In which country was he born?

ANSWER In Khorezm, in present-day Uzbekistan

Hard · stretch and reason

7 PROBLEMS

1. Write $7x - 5 = 3x + 11$ as a medieval sentence

ANSWER "Seven things less five dirhams equal three things and eleven dirhams"

2. Solve it, naming both moves

ANSWER Al-jabr, then al-muqabala; $x = 4$

3. Why do the two moves suffice for any linear equation?

ANSWER One removes subtractions, the other removes repetitions

4. What does the modern phrase "change the sign" correspond to?

ANSWER Al-jabr

5. What does "collect like terms" correspond to?

ANSWER Al-muqabala

6. Why did he prove $(x + 3)^2$ with a diagram?

ANSWER There was no symbolic algebra to argue with

7. Name the two words modern mathematics owes him

ANSWER Algebra and algorithm

07 Homework

Homework — 5 tasks

Name the move beside every line, as he would have done.

1. Solve $6x - 4 = 3x + 11$, naming al-jabr and al-muqabala where you use them.
2. Write "six things less one dirham equal three things and eight dirhams" as an equation and solve it.
3. Explain in three sentences why algebra is named after a book written in Baghdad.
4. Write $9x + 2 = 4x + 22$ as a medieval sentence.

5. Find out one more contribution of Al-Khwarizmi and write two sentences about it.